

DATA 266 Homework 3

DATA 266 — Homework 3

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Same SID/SEED as HW1. No extra HP_ID network; unmasked and masked attention are the two trained models. Notebook: `attention.ipynb`.

1. Prompt engineering

I called the model from code with LangChain PromptTemplate (not chat screenshots). Same local Flan-T5-small I used in HW2. Two prompts per technique: 15% of 80 (answer is 12) and a text/logic one.

What differed

Zero-shot already got the river-bank sense right (`side of a river`) but failed the math (98.6). Few-shot kept the bank answer (`river`) and still missed the percent (59). CoT copied 80% instead of computing 12. Zero-shot CoT was worse: it said the answer was 15, and picked Bob as youngest. Meta-prompting never produced a prompt (80% then a sentence about population). Tree of Thoughts printed -2.5 and echoed the telescope sentence.

On this model, extra prompt structure did not fix arithmetic. Few-shot was the only one that stayed short and still correct on the wording task. The longer templates mostly added junk. I still ran all 12 snippets in code like the assignment asked.

Full table is in `METRICS.md`.

2. Self-attention

Word-level tokens on the exact 5-sentence paragraph (46 tokens, vocab 40).
nn.Embedding + Q, K, V linears. Scaled dot-product $q \cdot k^T / \sqrt{d}$, softmax, times V. No nn.MultiheadAttention. Next-token cross entropy, seed 670, 400 epochs, Adam 0.01.

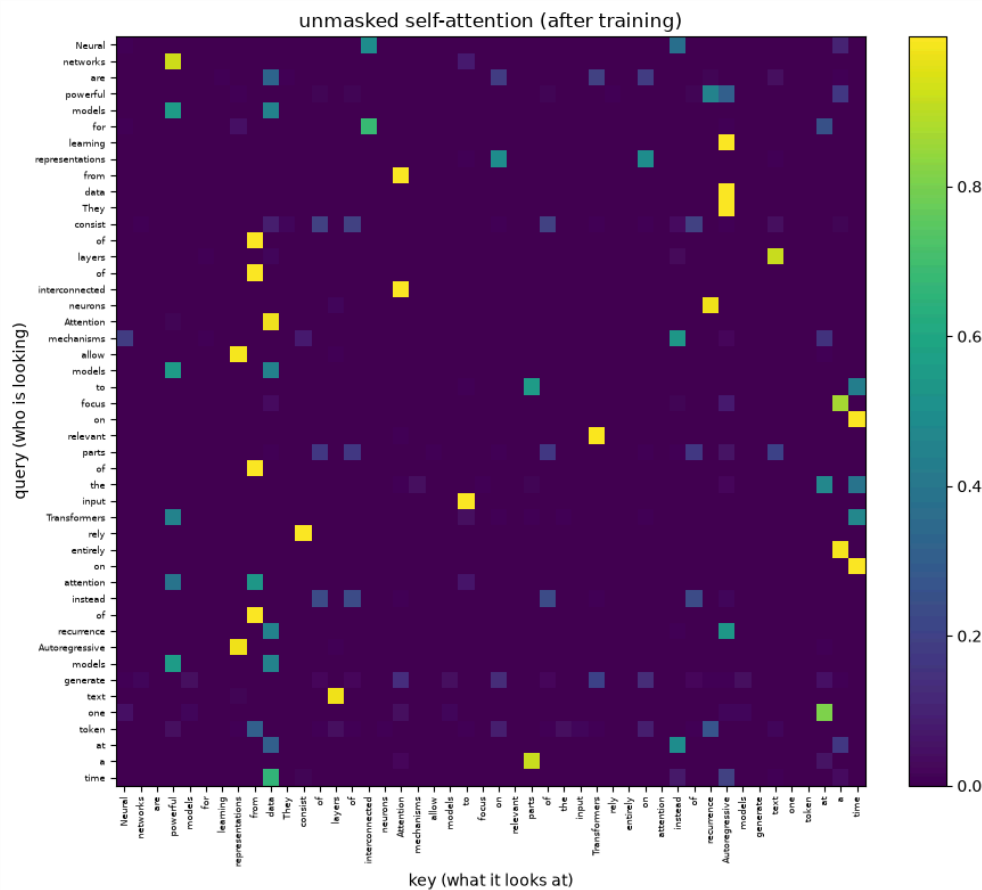
Unmasked: every token can look at every token. Loss ~ 0.33 . Row sums are 1.

Causal: future scores set to $-\infty$ **before** softmax. Upper triangle max was 0.0.
Loss got down around 0.06 then 0.21 at epoch 400.

Heatmaps

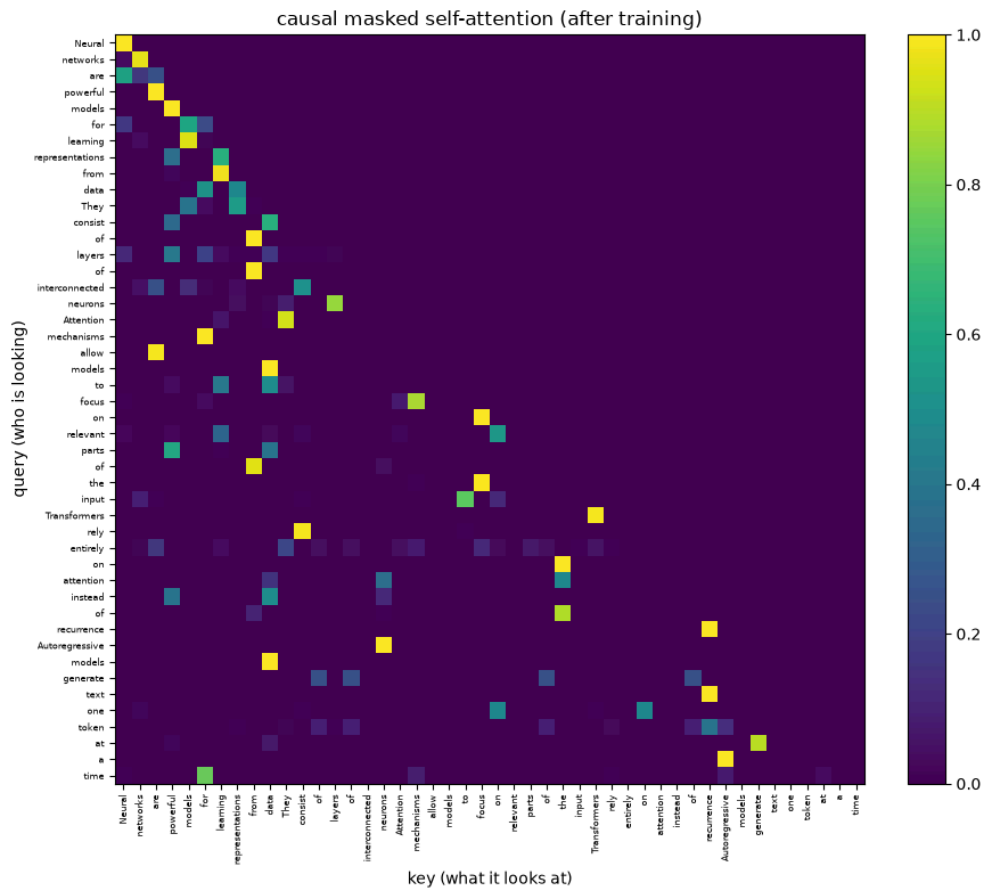
Taken from the executed notebook cells (not a phone screenshot).

Unmasked self-attention (after training)



unmasked attention heatmap

Causal masked self-attention (after training)



causal attention heatmap

Compare

Unmasked still puts weight on later words (bright cells to the right of the diagonal). That is allowed because there is no mask. Masked is lower triangular and the upper triangle is 0, so it cannot look ahead. That is what you want if you generate one token at a time. The two heatmaps are the two trained models for this assignment.